



TaskPilot

AI Project Management Chat Module



Technical Report

Functional Documentation & System Design

Platform

Stack

AI

Event

Version

🏆 COMPETITION SUBMISSION READY



Chat



Tasks



Dashboard



Team

Abstract

Executive Summary

TaskPilot is a production-quality **AI Project Management Chat Module**: conversational control, structured task CRUD, explainable priority scoring (0–100), live progress analytics, and team slots (max. 8). It runs entirely on **localhost** with Flask + SQLite and a polished dark-themed GUI. Built-in AI guarantees reliable commands; optional GPT enriches natural-language replies.

5

Core
mod-
ules

12+

REST
end-
points

0–
100

Priority
score

8

Team
slots

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Introduction

“The best interface is a conversation that gets work done.”

— TaskPilot design principle

Vision

Modern teams need more than static todo lists. **TaskPilot** unifies *natural language*, *structured tasks*, and *visual analytics* so users plan smarter and ship faster.

Problem: Manual prioritization wastes time and hides deadline risk.

Solution: AI-assisted creation, automatic scoring, and real-time dashboards.

Delivery: Full web product at <http://127.0.0.1:5000/login/> with session auth and REST API.

Core Objectives

1. **Chat-first workflow** — create, list, and prioritize via conversation
2. **Transparent AI** — every score is explainable to judges and users
3. **Production GUI** — not a CLI-only prototype; complete visual product
4. **Submission-ready** — README, `.env.example`, docs, share ZIP, this report

Functional Requirements

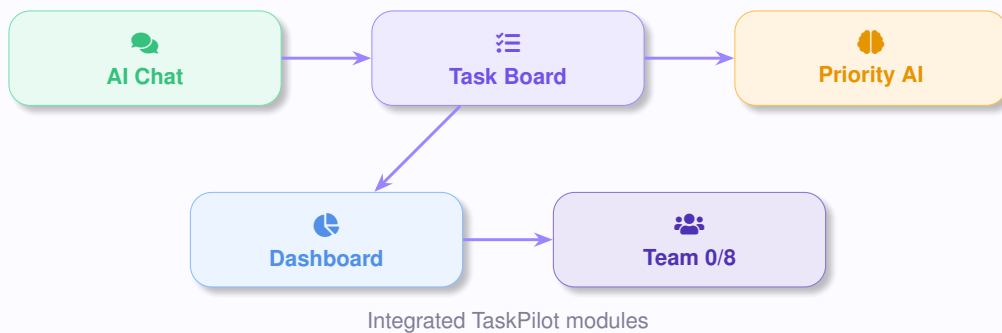


Figure 2.1: Functional module map

Requirements Traceability

Table 2.1: Competition requirements mapped to TaskPilot features

Requirement	Implementation
Chat interface	AI panel, history, quick chips, commands (add task, dashboard, top priorities)
Task management	CRUD, search, filters, modal editor, deadline quick-picks, priority pills
Priority scoring	Multi-factor 0–100 score; Critical / High / Medium / Low + explain API
Progress dashboard	Completion %, donut chart, status bars, priority grid, top-8 queue
Team collaboration	Up to 8 members, roles, live N / 8 Occupied counter
Localhost GUI	Flask + session cookies; dark theme; Demo Data one-click samples

System Architecture

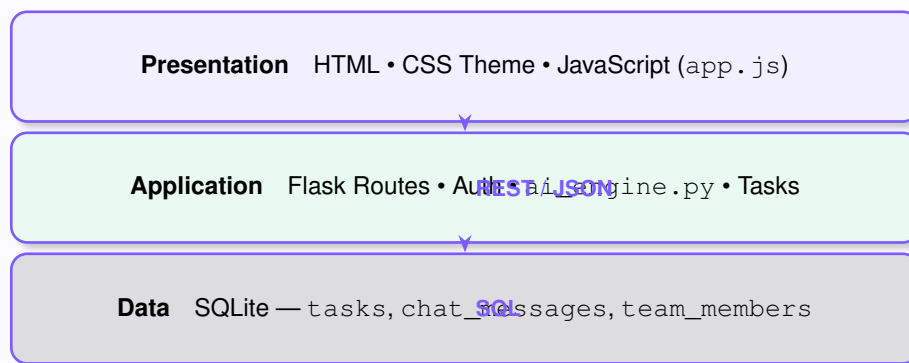
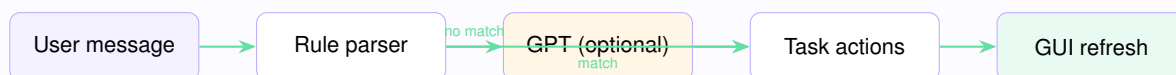


Figure 3.1: Three-tier system architecture

AI Message Pipeline



User Interface & Features

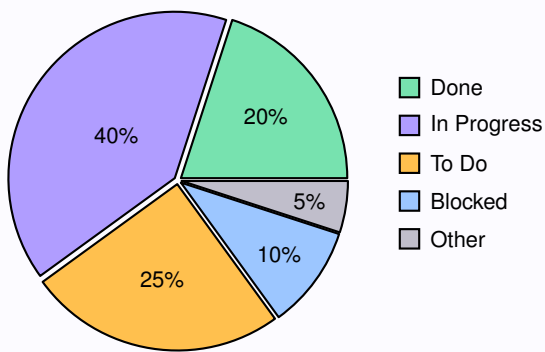


Figure 4.1: Sample task status mix (demo data)

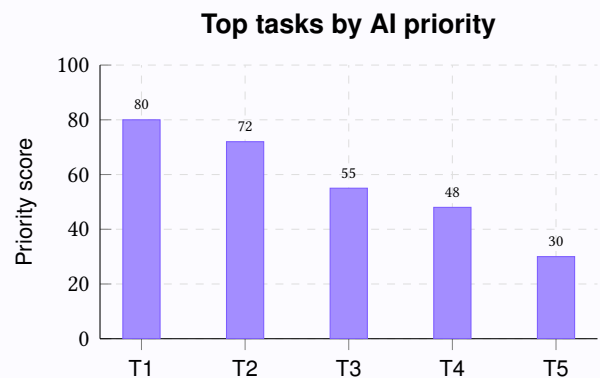


Figure 4.2: Priority score distribution

[

Login & Auth] Terminal-styled sign-in, demo mode (any email/password), session cookies, logout.

[

Welcome Hero] Personalized *Welcome back, Name!* banner with purple gradient highlight.

Workspace Highlights

- **AI Assistant** – GPT / Built-in badge, connection status, typing indicator
- **Task Board** – priority-sorted cards, edit/delete, *Explain priority*
- **Toolbar** – **Demo Data** (replace samples), Recalculate, AI Suggest Deadlines

Dashboard & Team

Dashboard shows completion %, stat cards, donut chart, status breakdown, priority grid, and top-8 active queue. Team tab supports roles (Member, Lead, Developer, Designer) with **8** maximum slots.

AI Priority Algorithm

Table 5.1: Priority bands (aligned with UI pills)

Score	Label	Visual
≥ 75	Critical	CRITICAL
55–74	High	HIGH
35–54	Medium	MEDIUM
< 35	Low	LOW

Scoring Model

$$\text{Score} = \text{clamp}(40 + w_{\text{kw}} + w_{\text{dl}} + w_{\text{st}} + w_{\text{ef}}, 0, 100)$$

Keywords (urgent, critical), deadline proximity, status, and effort hours drive the result.

API Reference

Method	Endpoint	Description
GET	/login/	Login page
GET	/	Main workspace (auth required)
POST	/api/chat	AI assistant message + actions
GET	/api/dashboard	Stats and chart payloads
GET/POST	/api/tasks	List or create tasks
PATCH	/api/tasks/<id>	Update task fields
DELETE	/api/tasks/<id>	Remove task
POST	/api/seed-demo	Load demo tasks (<code>replace: true</code>)
GET/POST	/api/team	Team members
POST	/api/tasks/recalculate	Re-score all tasks

Table 6.1: Primary REST API surface

Installation & Configuration

>_ Quick Start — Windows

```
1 python -m venv venv
2 .\venv\Scripts\activate
3 pip install -r requirements.txt
4 copy .env.example .env
5 python run.py
6 # Browser: http://127.0.0.1:5000/login/
```

Note

OpenAI is **optional**. All task commands work with the built-in rule engine; set `OPENAI_API_KEY` only for richer chat replies.

Testing & Judge Demo

Automated Verification

`scripts/test_api.py` exercises health, login, seed-demo, chat task creation, priorities, and dashboard.

Manual Checklist

- ✓ Login + welcome banner
- ✓ **Demo Data** loads 5 sample tasks
- ✓ Dashboard charts populate
- ✓ Chat: add task ... urgent due tomorrow
- ✓ Deadline quick-picks and clear
- ✓ Team add/remove within 8 slots

5-Minute Demo Script

1. Problem statement (30 s)
2. Chat commands + **Demo Data** (90 s)
3. Task board + AI suggest deadlines (60 s)
4. Dashboard analytics tour (60 s)
5. Team tab + wrap-up (60 s)

Conclusion

TaskPilot is a complete, judge-ready submission: working localhost GUI, Flask backend, explainable AI priorities, professional documentation, and distributable `TaskPilot-share.zip`.

Strengths: Reliability, visual polish, modular Python packages, full competition checklist.

Future enhancements: Cloud hosting, WebSockets, Kanban lanes, email notifications.

Thank you for reviewing TaskPilot

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